

# CO3 AT2

## Level 2 Exam

NAME: A K Selva Prabhu

REG NO: 192421224

COURSE CODE: CSA0904

COURSE NAME: PROGRAMMING FOR JAVA

1. Write a Program to Find the Nth Largest Number in a array. Sample Input: List : {14, 67, 48, 23, 5, 62} N = 4 Sample Output: 4 th Largest number: 23

```
import java.util.Arrays;
```

```
import java.util.Scanner;
```

```
class Main {
```

```
    public static void main(String[] args) {
```

```
        int[] arr = {14, 67, 48, 23, 5, 62};
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter N: ");
```

```
        int n = sc.nextInt();
```

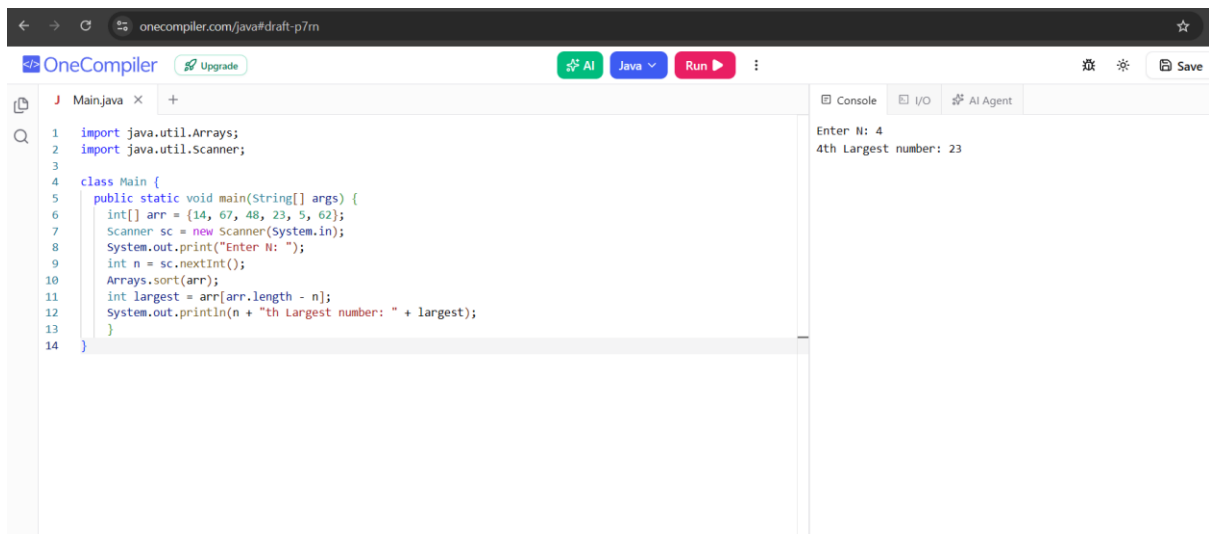
```
        Arrays.sort(arr);
```

```
        int largest = arr[arr.length - n];
```

```
        System.out.println(n + "th Largest number: " + largest);
```

```
    }
```

}



The screenshot shows the OneCompiler web IDE. The code editor contains a Java program that finds the Nth largest number in an array. The console output shows the user entered 4, and the program output is "4th Largest number: 23".

```
1 import java.util.Arrays;
2 import java.util.Scanner;
3
4 class Main {
5     public static void main(String[] args) {
6         int[] arr = {14, 67, 48, 23, 5, 62};
7         Scanner sc = new Scanner(System.in);
8         System.out.print("Enter N: ");
9         int n = sc.nextInt();
10        Arrays.sort(arr);
11        int largest = arr[arr.length - n];
12        System.out.println(n + "th Largest number: " + largest);
13    }
14 }
```

Console Output:

```
Enter N: 4
4th Largest number: 23
```

2 Find the count of prime numbers and composite numbers among the given inputs

```
import java.util.Scanner;
```

```
class Main {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int primeCount = 0;
```

```
        int compositeCount = 0;
```

```
        System.out.print("Enter how many numbers: ");
```

```
        int n = sc.nextInt();
```

```
        for (int i = 1; i <= n; i++) {
```

```
            System.out.print("Enter number: ");
```

```
int num = sc.nextInt();
```

```
int count = 0;
```

```
for (int j = 1; j <= num; j++) {
```

```
    if (num % j == 0) {
```

```
        count++;
```

```
    }
```

```
}
```

```
if (count == 2) {
```

```
    primeCount++;
```

```
} else if (count > 2) {
```

```
    compositeCount++;
```

```
}
```

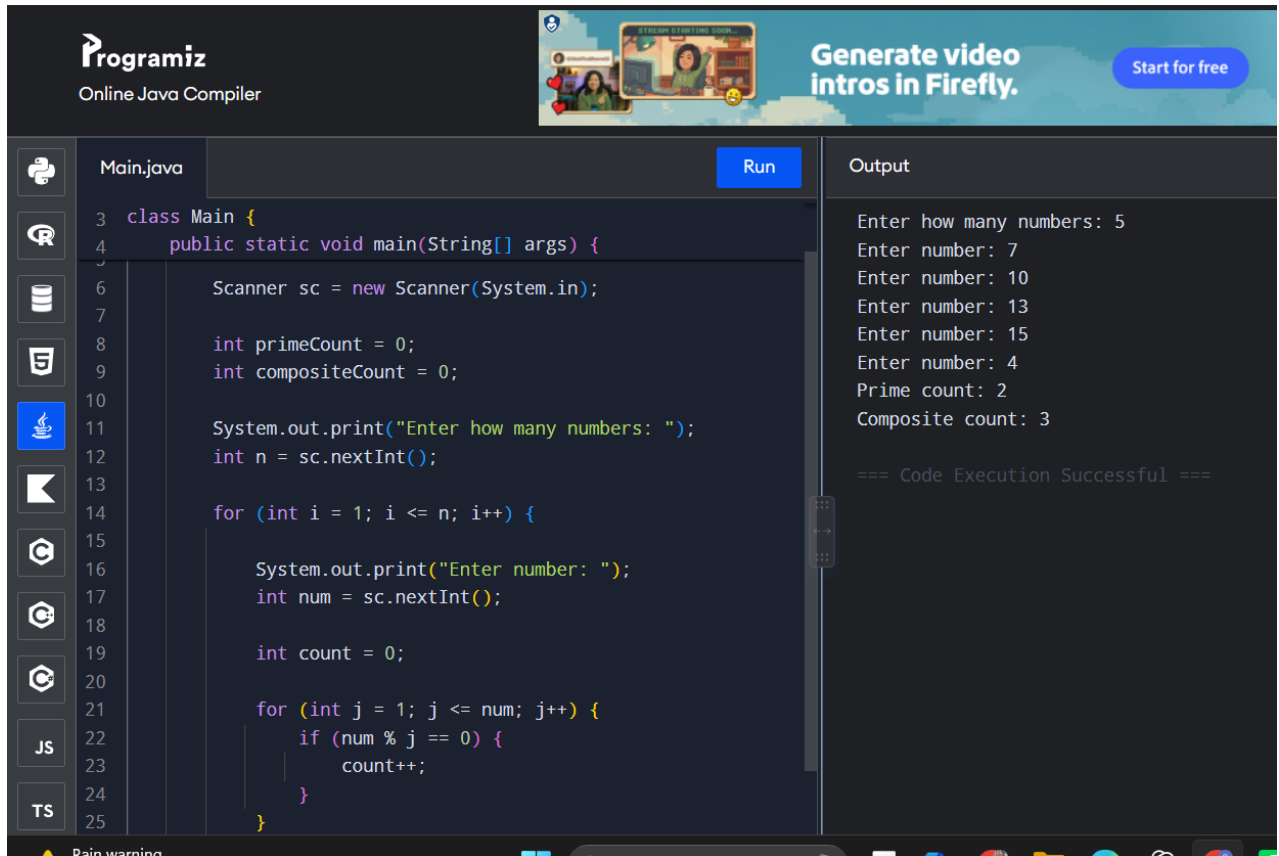
```
}
```

```
System.out.println("Prime count: " + primeCount);
```

```
System.out.println("Composite count: " + compositeCount);
```

```
}
```

```
}
```



3 In an organization they decide to give bonus to all the employees on New Year. A 5% bonus on salary is given to the grade A workers and 10% bonus on salary to the grade B workers. Write a program to enter the salary and grade of the employee. If the salary of the employee is less than \$10,000 then the employee gets an extra 2% bonus on salary Calculate the bonus that has to be given to the employee and print the salary that the employee will get. Sample Input & Output: Enter the grade of the employee: B Enter the employee salary: 50000 Salary=50000 Bonus=5000.0 Total to be paid:55000.0 give java code simple java code

```
import java.util.Scanner;
```

```
class Main {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter the grade of the employee: ");
```

```
char grade = sc.next().charAt(0);
```

```
System.out.print("Enter the employee salary: ");
```

```
double salary = sc.nextDouble();
```

```
double bonus = 0;
```

```
if (grade == 'A') {
```

```
    bonus = salary * 0.05;
```

```
} else if (grade == 'B') {
```

```
    bonus = salary * 0.10;
```

```
}
```

```
if (salary < 10000) {
```

```
    bonus = bonus + salary * 0.02;
```

```
}
```

```
double total = salary + bonus;
```

```
System.out.println("Salary=" + salary);
```

```
System.out.println("Bonus=" + bonus);
```

```
System.out.println("Total to be paid:" + total);
```

```
}
```

```
}
```

| Main.java                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Run | Output                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>1 import java.util.Scanner; 2 3 class Main { 4     public static void main(String[] args) { 5 6         Scanner sc = new Scanner(System.in); 7 8         System.out.print("Enter the grade of the employee: "); 9         char grade = sc.next().charAt(0); 10 11        System.out.print("Enter the employee salary: "); 12        double salary = sc.nextDouble(); 13 14        double bonus = 0; 15 16        if (grade == 'A') { 17            bonus = salary * 0.05; 18        } else if (grade == 'B') { 19            bonus = salary * 0.10; 20        } 21 22        if (salary &lt; 10000) { 23            bonus = bonus + salary * 0.02;</pre> |     | <pre>Enter the grade of the employee: b Enter the employee salary: 50000 Salary=50000.0 Bonus=0.0 Total to be paid:50000.0  === Code Execution Successful ===</pre> |